Retrieval-Augmented Generation: A Short Study

Retrieval-Augmented Generation (RAG) lets a language model read external documents at question time. Because the knowledge lives in the documents and not in the model weights, it can be updated without retraining. The main risk is that a wrong chunk leads to a wrong answer.

2. Models and Results

We compared three models on the same test set of 2,000 questions.
Model | Accuracy | Parameters
BERT-base | 88.4% | 110M
RoBERTa-large | 91.2% | 355M
DistilBERT | 85.1% | 66M
The best model, RoBERTa-large, is also the
largest, which makes it slower at inference time.

3. Scoring

The final score of each model is computed with Equation 12:
score = 0.7 * accuracy + 0.3 * speed
where speed is the number of questions answered per second, divided by the speed of the fastest model.